

Biomedical Data Processing: final project report

ECG-based identification and EEG-based sleep staging

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1 Part I: ECG-based identification

The aim of this part is to recover, without supervision, which of five subjects produced each heartbeat in a single continuous ECG recording. The pipeline has six stages: R-peak detection with Pan-Tompkins, Wiener filtering (a PQRST model as the desired signal, the inter-beat intervals as the noise), segmentation into one-beat windows, wavelet decomposition, *K*-means clustering into five groups, and PCA followed by re-clustering to measure the cost of the compression. The ground-truth labels are never used inside this chain, only to score the finished partition, and since *K*-means numbers its clusters arbitrarily, the labelling is resolved afterwards by searching over all permutations of the five indices.

1.1 Data exploration

The recording is a single ECG channel sampled at 200 Hz, 605 s long, and contains the heartbeats of five subjects who touch the sensor in turn, one beat at a time. Figure 1 shows a 20 s window of the unfiltered signal.

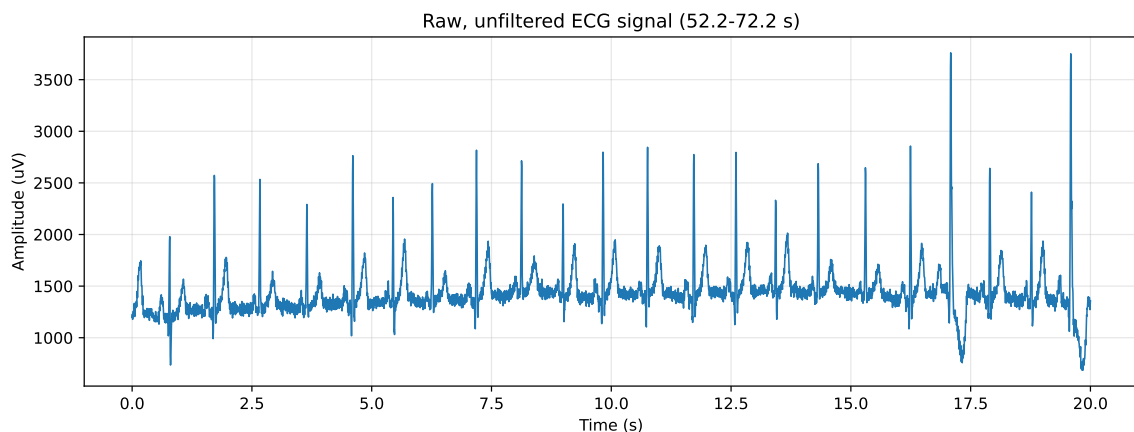


Figure 1: Twenty seconds of the raw, unfiltered ECG recording (52.2–72.2 s). The window contains 22 beats, an average rate of about 66 bpm, and all five subjects contribute to it.

Which noise sources are present, and why? Three sources are visible directly on this plot.

Baseline wander.

The isoelectric line drifts between roughly 1210 μV and 1550 μV , a swing of 340 μV , far too slow to be cardiac activity. It tracks the electrode-skin contact, which changes with each subject: the deep dip to 757 μV near $t = 17\text{ s}$ coincides with a subject change and has no PQRST shape.

Powerline interference.

A regular ripple of constant amplitude is present everywhere, including the flat segments between beats where the heart contributes nothing. Isolated around 50 Hz it has an rms of 27.9 μV and a peak-to-peak amplitude of about 104 μV ; its periodicity and its indifference to whether a beat is present identify it as mains interference rather than biology.

Broadband muscle and contact noise.

An irregular, burst-like fuzz sits on top of the ripple, best seen on the flat segments. The residual

above 60 Hz, well outside the ECG band, has an rms of $14.9 \mu\text{V}$ and is non-periodic with intensity that varies over time, consistent with muscle contractions and electrode micro-movement.

1.1.1 Confirmation in the frequency domain

As an optional addition I estimated the power spectral density of the whole recording with Welch's method (Hann window, 4096 samples, 50 % overlap) to confirm the three sources quantitatively. Figure 2 shows the result.

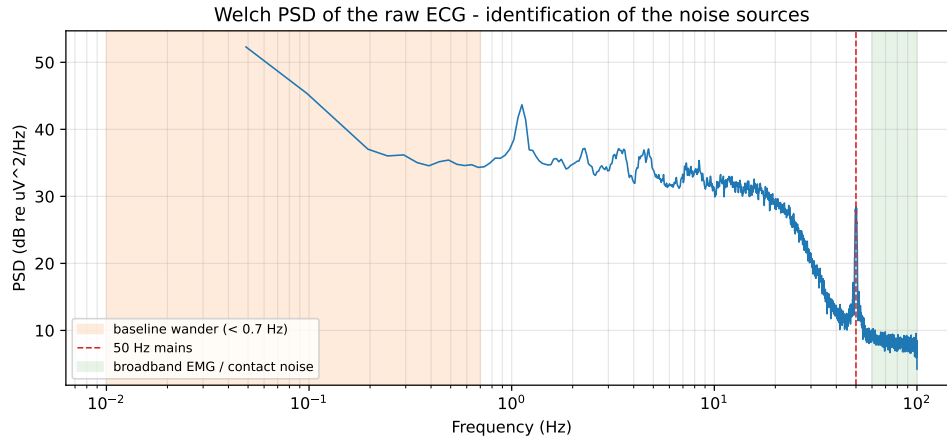


Figure 2: Welch power spectral density of the complete raw recording.

The band below 0.7 Hz carries 22.3 % of the total power, confirming the baseline wander as the dominant disturbance. A narrow line stands 15.1 dB above the floor at exactly 50 Hz with no line at 60 Hz and no harmonic at 100 Hz, confirming a clean mains tone. Above 60 Hz, well past where the ECG itself carries content, the floor stays elevated, confirming the broadband muscle and contact noise. The three contributions occupy fixed bands and do not change character along the record, so the noise is stationary, which is what justifies the single, time-invariant Wiener filter of Section 1.2.

1.2 Wiener filter

To be written.